

REM2 Research Methods

Exercise 4-1: Rental vs. Sales Markets

Instructions: Work in groups of two to complete the following tasks

1. Open “Lectures 4-6 Exercises Data.xlsx” in a software of your choice. The file contains the following variables.

Variable name	Definition
Transtype	Sales or rental transaction
Propertytype	Property type (i.e., flat/apartment, detached, terraced, semi-detached, or bungalow)
Date	Date of transaction
Price	Selling or rental price in £
Floorarea	Floor area in square metres
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
Busstation	Number of bus stations within a 0.5-mile radius of the estate
Railstation	Shortest distance to a train station in metres
Airport	Shortest distance to an airport in metres

2. Generate descriptive statistics of all variables. Note the distribution of ‘Floorarea’. What is the maximum value of this variable? Remove observations with a floor area over 200 squared metres.
3. Make natural log transformation of the following variables: Price, Floorarea, Busstation, Railstation and Airport. Save the file to your work directory.
4. Convert Propertytype and Date into dummy variables.
5. Estimate a hedonic price model for the rental and the sales market, respectively. Include intercept dummies for property type and transaction year. Use both the linear and the log-linear functional form, as outlined below.

Linear functional form:

$$\begin{aligned} \text{Price} = & \beta_0 + \beta_1 \text{Floorarea} + \beta_2 \text{Busstation} + \beta_3 \text{Railstation} + \beta_4 \text{Airport} \\ & + \beta_5 \text{Bedrooms} + \beta_6 \text{Bathrooms} + \beta_7 \text{Terraced House} + \beta_8 \text{Semi} \\ & - \text{Detached House} + \beta_9 \text{Bungalow} + \beta_{10} \text{Detached House} \\ & + \beta_{11} \text{Year}_{2018} + \beta_{12} \text{Year}_{2019} + \beta_{13} \text{Year}_{2020} + \beta_{14} \text{Year}_{2021} \\ & + \beta_{15} \text{Year}_{2022} + \varepsilon \end{aligned}$$

Log-linear functional form:

$$\begin{aligned}
\ln(\text{Price}) = & \beta_0 + \beta_1 \ln(\text{Floorarea}) + \beta_2 \ln(\text{Busstation}) + \beta_3 \ln(\text{Railstation}) \\
& + \beta_4 \text{Airport} + \beta_5 \text{Bedrooms} + \beta_6 \text{Bathrooms} \\
& + \beta_7 \text{Terraced House} + \beta_8 \text{Semi-Detached House} \\
& + \beta_9 \text{Bungalow} + \beta_{10} \text{Detached House} + \beta_{11} \text{Year}_{2018} \\
& + \beta_{12} \text{Year}_{2019} + \beta_{13} \text{Year}_{2020} + \beta_{14} \text{Year}_{2021} + \beta_{15} \text{Year}_{2022} + \varepsilon
\end{aligned}$$

6. Estimate a hedonic price model for the rental market only. Include a slope dummy to check if the effect of Floorarea is different between Detached houses and other types of properties.